DLCV HW1

Problem 1

1.

Model: Resnet152

```
----------------------------------------------------------------
       Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1          [-1, 64, 48, 48]           9,408
       BatchNorm2d-2          [-1, 64, 48, 48]             128
              ReLU-3          [-1, 64, 48, 48]               0
         MaxPool2d-4          [-1, 64, 24, 24]               0
            Conv2d-5          [-1, 64, 24, 24]           4,096
       BatchNorm2d-6          [-1, 64, 24, 24]             128
              ReLU-7          [-1, 64, 24, 24]               0
            Conv2d-8          [-1, 64, 24, 24]          36,864
       BatchNorm2d-9          [-1, 64, 24, 24]             128
             ReLU-10          [-1, 64, 24, 24]               0
           Conv2d-11         [-1, 256, 24, 24]          16,384
      BatchNorm2d-12         [-1, 256, 24, 24]             512
           Conv2d-13         [-1, 256, 24, 24]          16,384
      BatchNorm2d-14         [-1, 256, 24, 24]             512
             ReLU-15         [-1, 256, 24, 24]               0
      Bottleneck-16          [-1, 256, 24, 24]               0
           Conv2d-17          [-1, 64, 24, 24]          16,384
      BatchNorm2d-18          [-1, 64, 24, 24]             128
             ReLU-19          [-1, 64, 24, 24]               0
           Conv2d-20          [-1, 64, 24, 24]          36,864
      BatchNorm2d-21          [-1, 64, 24, 24]             128
             ReLU-22          [-1, 64, 24, 24]               0
           Conv2d-23         [-1, 256, 24, 24]          16,384
      BatchNorm2d-24         [-1, 256, 24, 24]             512
             ReLU-25         [-1, 256, 24, 24]               0
      Bottleneck-26          [-1, 256, 24, 24]               0
           Conv2d-27          [-1, 64, 24, 24]          16,384
      BatchNorm2d-28          [-1, 64, 24, 24]             128
```

ReLU-29	[-1, 64, 24, 24]	0
Conv2d-30	[-1, 64, 24, 24]	36,864
BatchNorm2d-31	[-1, 64, 24, 24]	128
ReLU-32	[-1, 64, 24, 24]	0
Conv2d-33	[-1, 256, 24, 24]	16,384
BatchNorm2d-34	[-1, 256, 24, 24]	512
ReLU-35	[-1, 256, 24, 24]	0
Bottleneck-36	[-1, 256, 24, 24]	0
Conv2d-37	[-1, 128, 24, 24]	32,768
BatchNorm2d-38	[-1, 128, 24, 24]	256
ReLU-39	[-1, 128, 24, 24]	0
Conv2d-40	[-1, 128, 12, 12]	147,456
BatchNorm2d-41	[-1, 128, 12, 12]	256
ReLU-42	[-1, 128, 12, 12]	0
Conv2d-43	[-1, 512, 12, 12]	65,536
BatchNorm2d-44	[-1, 512, 12, 12]	1,024
Conv2d-45	[-1, 512, 12, 12]	131,072
BatchNorm2d-46	[-1, 512, 12, 12]	1,024
ReLU-47	[-1, 512, 12, 12]	0
Bottleneck-48	[-1, 512, 12, 12]	0
Conv2d-49	[-1, 128, 12, 12]	65,536
BatchNorm2d-50	[-1, 128, 12, 12]	256
ReLU-51	[-1, 128, 12, 12]	0
Conv2d-52	[-1, 128, 12, 12]	147,456
BatchNorm2d-53	[-1, 128, 12, 12]	256
ReLU-54	[-1, 128, 12, 12]	0
Conv2d-55	[-1, 512, 12, 12]	65,536
BatchNorm2d-56	[-1, 512, 12, 12]	1,024
ReLU-57	[-1, 512, 12, 12]	0
Bottleneck-58	[-1, 512, 12, 12]	0
Conv2d-59	[-1, 128, 12, 12]	65,536
BatchNorm2d-60	[-1, 128, 12, 12]	256
ReLU-61	[-1, 128, 12, 12]	0
Conv2d-62	[-1, 128, 12, 12]	147,456
BatchNorm2d-63	[-1, 128, 12, 12]	256
ReLU-64	[-1, 128, 12, 12]	0
Conv2d-65	[-1, 512, 12, 12]	65,536
BatchNorm2d-66	[-1, 512, 12, 12]	1,024

ReLU-67	[-1, 512, 12, 12]	0
Bottleneck-68	[-1, 512, 12, 12]	0
Conv2d-69	[-1, 128, 12, 12]	65,536
BatchNorm2d-70	[-1, 128, 12, 12]	256
ReLU-71	[-1, 128, 12, 12]	0
Conv2d-72	[-1, 128, 12, 12]	147,456
BatchNorm2d-73	[-1, 128, 12, 12]	256
ReLU-74	[-1, 128, 12, 12]	0
Conv2d-75	[-1, 512, 12, 12]	65,536
BatchNorm2d-76	[-1, 512, 12, 12]	1,024
ReLU-77	[-1, 512, 12, 12]	0
Bottleneck-78	[-1, 512, 12, 12]	0
Conv2d-79	[-1, 128, 12, 12]	65,536
BatchNorm2d-80	[-1, 128, 12, 12]	256
ReLU-81	[-1, 128, 12, 12]	0
Conv2d-82	[-1, 128, 12, 12]	147,456
BatchNorm2d-83	[-1, 128, 12, 12]	256
ReLU-84	[-1, 128, 12, 12]	0
Conv2d-85	[-1, 512, 12, 12]	65,536
BatchNorm2d-86	[-1, 512, 12, 12]	1,024
ReLU-87	[-1, 512, 12, 12]	0
Bottleneck-88	[-1, 512, 12, 12]	0
Conv2d-89	[-1, 128, 12, 12]	65,536
BatchNorm2d-90	[-1, 128, 12, 12]	256
ReLU-91	[-1, 128, 12, 12]	0
Conv2d-92	[-1, 128, 12, 12]	147,456
BatchNorm2d-93	[-1, 128, 12, 12]	256
ReLU-94	[-1, 128, 12, 12]	0
Conv2d-95	[-1, 512, 12, 12]	65,536
BatchNorm2d-96	[-1, 512, 12, 12]	1,024
ReLU-97	[-1, 512, 12, 12]	0
Bottleneck-98	[-1, 512, 12, 12]	0
Conv2d-99	[-1, 128, 12, 12]	65,536
BatchNorm2d-100	[-1, 128, 12, 12]	256
ReLU-101	[-1, 128, 12, 12]	0
Conv2d-102	[-1, 128, 12, 12]	147,456
BatchNorm2d-103	[-1, 128, 12, 12]	256
ReLU-104	[-1, 128, 12, 12]	0

Conv2d-105	[-1, 512, 12, 12]	65,536
BatchNorm2d-106	[-1, 512, 12, 12]	1,024
ReLU-107	[-1, 512, 12, 12]	0
Bottleneck-108	[-1, 512, 12, 12]	0
Conv2d-109	[-1, 128, 12, 12]	65,536
BatchNorm2d-110	[-1, 128, 12, 12]	256
ReLU-111	[-1, 128, 12, 12]	0
Conv2d-112	[-1, 128, 12, 12]	147,456
BatchNorm2d-113	[-1, 128, 12, 12]	256
ReLU-114	[-1, 128, 12, 12]	0
Conv2d-115	[-1, 512, 12, 12]	65,536
BatchNorm2d-116	[-1, 512, 12, 12]	1,024
ReLU-117	[-1, 512, 12, 12]	0
Bottleneck-118	[-1, 512, 12, 12]	0
Conv2d-119	[-1, 256, 12, 12]	131,072
BatchNorm2d-120	[-1, 256, 12, 12]	512
ReLU-121	[-1, 256, 12, 12]	0
Conv2d-122	[-1, 256, 6, 6]	589,824
BatchNorm2d-123	[-1, 256, 6, 6]	512
ReLU-124	[-1, 256, 6, 6]	0
Conv2d-125	[-1, 1024, 6, 6]	262,144
BatchNorm2d-126	[-1, 1024, 6, 6]	2,048
Conv2d-127	[-1, 1024, 6, 6]	524,288
BatchNorm2d-128	[-1, 1024, 6, 6]	2,048
ReLU-129	[-1, 1024, 6, 6]	0
Bottleneck-130	[-1, 1024, 6, 6]	0
Conv2d-131	[-1, 256, 6, 6]	262,144
BatchNorm2d-132	[-1, 256, 6, 6]	512
ReLU-133	[-1, 256, 6, 6]	0
Conv2d-134	[-1, 256, 6, 6]	589,824
BatchNorm2d-135	[-1, 256, 6, 6]	512
ReLU-136	[-1, 256, 6, 6]	0
Conv2d-137	[-1, 1024, 6, 6]	262,144
BatchNorm2d-138	[-1, 1024, 6, 6]	2,048
ReLU-139	[-1, 1024, 6, 6]	0
Bottleneck-140	[-1, 1024, 6, 6]	0
Conv2d-141	[-1, 256, 6, 6]	262,144
BatchNorm2d-142	[-1, 256, 6, 6]	512

ReLU-143	[-1, 256, 6, 6]	0
Conv2d-144	[-1, 256, 6, 6]	589,824
BatchNorm2d-145	[-1, 256, 6, 6]	512
ReLU-146	[-1, 256, 6, 6]	0
Conv2d-147	[-1, 1024, 6, 6]	262,144
BatchNorm2d-148	[-1, 1024, 6, 6]	2,048
ReLU-149	[-1, 1024, 6, 6]	0
Bottleneck-150	[-1, 1024, 6, 6]	0
Conv2d-151	[-1, 256, 6, 6]	262,144
BatchNorm2d-152	[-1, 256, 6, 6]	512
ReLU-153	[-1, 256, 6, 6]	0
Conv2d-154	[-1, 256, 6, 6]	589,824
BatchNorm2d-155	[-1, 256, 6, 6]	512
ReLU-156	[-1, 256, 6, 6]	0
Conv2d-157	[-1, 1024, 6, 6]	262,144
BatchNorm2d-158	[-1, 1024, 6, 6]	2,048
ReLU-159	[-1, 1024, 6, 6]	0
Bottleneck-160	[-1, 1024, 6, 6]	0
Conv2d-161	[-1, 256, 6, 6]	262,144
BatchNorm2d-162	[-1, 256, 6, 6]	512
ReLU-163	[-1, 256, 6, 6]	0
Conv2d-164	[-1, 256, 6, 6]	589,824
BatchNorm2d-165	[-1, 256, 6, 6]	512
ReLU-166	[-1, 256, 6, 6]	0
Conv2d-167	[-1, 1024, 6, 6]	262,144
BatchNorm2d-168	[-1, 1024, 6, 6]	2,048
ReLU-169	[-1, 1024, 6, 6]	0
Bottleneck-170	[-1, 1024, 6, 6]	0
Conv2d-171	[-1, 256, 6, 6]	262,144
BatchNorm2d-172	[-1, 256, 6, 6]	512
ReLU-173	[-1, 256, 6, 6]	0
Conv2d-174	[-1, 256, 6, 6]	589,824
BatchNorm2d-175	[-1, 256, 6, 6]	512
ReLU-176	[-1, 256, 6, 6]	0
Conv2d-177	[-1, 1024, 6, 6]	262,144
BatchNorm2d-178	[-1, 1024, 6, 6]	2,048
ReLU-179	[-1, 1024, 6, 6]	0
Bottleneck-180	[-1, 1024, 6, 6]	0

Conv2d-181	[-1, 256, 6, 6]	262,144
BatchNorm2d-182	[-1, 256, 6, 6]	512
ReLU-183	[-1, 256, 6, 6]	0
Conv2d-184	[-1, 256, 6, 6]	589,824
BatchNorm2d-185	[-1, 256, 6, 6]	512
ReLU-186	[-1, 256, 6, 6]	0
Conv2d-187	[-1, 1024, 6, 6]	262,144
BatchNorm2d-188	[-1, 1024, 6, 6]	2,048
ReLU-189	[-1, 1024, 6, 6]	0
Bottleneck-190	[-1, 1024, 6, 6]	0
Conv2d-191	[-1, 256, 6, 6]	262,144
BatchNorm2d-192	[-1, 256, 6, 6]	512
ReLU-193	[-1, 256, 6, 6]	0
Conv2d-194	[-1, 256, 6, 6]	589,824
BatchNorm2d-195	[-1, 256, 6, 6]	512
ReLU-196	[-1, 256, 6, 6]	0
Conv2d-197	[-1, 1024, 6, 6]	262,144
BatchNorm2d-198	[-1, 1024, 6, 6]	2,048
ReLU-199	[-1, 1024, 6, 6]	0
Bottleneck-200	[-1, 1024, 6, 6]	0
Conv2d-201	[-1, 256, 6, 6]	262,144
BatchNorm2d-202	[-1, 256, 6, 6]	512
ReLU-203	[-1, 256, 6, 6]	0
Conv2d-204	[-1, 256, 6, 6]	589,824
BatchNorm2d-205	[-1, 256, 6, 6]	512
ReLU-206	[-1, 256, 6, 6]	0
Conv2d-207	[-1, 1024, 6, 6]	262,144
BatchNorm2d-208	[-1, 1024, 6, 6]	2,048
ReLU-209	[-1, 1024, 6, 6]	0
Bottleneck-210	[-1, 1024, 6, 6]	0
Conv2d-211	[-1, 256, 6, 6]	262,144
BatchNorm2d-212	[-1, 256, 6, 6]	512
ReLU-213	[-1, 256, 6, 6]	0
Conv2d-214	[-1, 256, 6, 6]	589,824
BatchNorm2d-215	[-1, 256, 6, 6]	512
ReLU-216	[-1, 256, 6, 6]	0
Conv2d-217	[-1, 1024, 6, 6]	262,144
BatchNorm2d-218	[-1, 1024, 6, 6]	2,048

ReLU-219	[-1, 1024, 6, 6]	0
Bottleneck-220	[-1, 1024, 6, 6]	0
Conv2d-221	[-1, 256, 6, 6]	262,144
BatchNorm2d-222	[-1, 256, 6, 6]	512
ReLU-223	[-1, 256, 6, 6]	0
Conv2d-224	[-1, 256, 6, 6]	589,824
BatchNorm2d-225	[-1, 256, 6, 6]	512
ReLU-226	[-1, 256, 6, 6]	0
Conv2d-227	[-1, 1024, 6, 6]	262,144
BatchNorm2d-228	[-1, 1024, 6, 6]	2,048
ReLU-229	[-1, 1024, 6, 6]	0
Bottleneck-230	[-1, 1024, 6, 6]	0
Conv2d-231	[-1, 256, 6, 6]	262,144
BatchNorm2d-232	[-1, 256, 6, 6]	512
ReLU-233	[-1, 256, 6, 6]	0
Conv2d-234	[-1, 256, 6, 6]	589,824
BatchNorm2d-235	[-1, 256, 6, 6]	512
ReLU-236	[-1, 256, 6, 6]	0
Conv2d-237	[-1, 1024, 6, 6]	262,144
BatchNorm2d-238	[-1, 1024, 6, 6]	2,048
ReLU-239	[-1, 1024, 6, 6]	0
Bottleneck-240	[-1, 1024, 6, 6]	0
Conv2d-241	[-1, 256, 6, 6]	262,144
BatchNorm2d-242	[-1, 256, 6, 6]	512
ReLU-243	[-1, 256, 6, 6]	0
Conv2d-244	[-1, 256, 6, 6]	589,824
BatchNorm2d-245	[-1, 256, 6, 6]	512
ReLU-246	[-1, 256, 6, 6]	0
Conv2d-247	[-1, 1024, 6, 6]	262,144
BatchNorm2d-248	[-1, 1024, 6, 6]	2,048
ReLU-249	[-1, 1024, 6, 6]	0
Bottleneck-250	[-1, 1024, 6, 6]	0
Conv2d-251	[-1, 256, 6, 6]	262,144
BatchNorm2d-252	[-1, 256, 6, 6]	512
ReLU-253	[-1, 256, 6, 6]	0
Conv2d-254	[-1, 256, 6, 6]	589,824
BatchNorm2d-255	[-1, 256, 6, 6]	512
ReLU-256	[-1, 256, 6, 6]	0

Conv2d-257	[-1, 1024, 6, 6]	262,144
BatchNorm2d-258	[-1, 1024, 6, 6]	2,048
ReLU-259	[-1, 1024, 6, 6]	0
Bottleneck-260	[-1, 1024, 6, 6]	0
Conv2d-261	[-1, 256, 6, 6]	262,144
BatchNorm2d-262	[-1, 256, 6, 6]	512
ReLU-263	[-1, 256, 6, 6]	0
Conv2d-264	[-1, 256, 6, 6]	589,824
BatchNorm2d-265	[-1, 256, 6, 6]	512
ReLU-266	[-1, 256, 6, 6]	0
Conv2d-267	[-1, 1024, 6, 6]	262,144
BatchNorm2d-268	[-1, 1024, 6, 6]	2,048
ReLU-269	[-1, 1024, 6, 6]	0
Bottleneck-270	[-1, 1024, 6, 6]	0
Conv2d-271	[-1, 256, 6, 6]	262,144
BatchNorm2d-272	[-1, 256, 6, 6]	512
ReLU-273	[-1, 256, 6, 6]	0
Conv2d-274	[-1, 256, 6, 6]	589,824
BatchNorm2d-275	[-1, 256, 6, 6]	512
ReLU-276	[-1, 256, 6, 6]	0
Conv2d-277	[-1, 1024, 6, 6]	262,144
BatchNorm2d-278	[-1, 1024, 6, 6]	2,048
ReLU-279	[-1, 1024, 6, 6]	0
Bottleneck-280	[-1, 1024, 6, 6]	0
Conv2d-281	[-1, 256, 6, 6]	262,144
BatchNorm2d-282	[-1, 256, 6, 6]	512
ReLU-283	[-1, 256, 6, 6]	0
Conv2d-284	[-1, 256, 6, 6]	589,824
BatchNorm2d-285	[-1, 256, 6, 6]	512
ReLU-286	[-1, 256, 6, 6]	0
Conv2d-287	[-1, 1024, 6, 6]	262,144
BatchNorm2d-288	[-1, 1024, 6, 6]	2,048
ReLU-289	[-1, 1024, 6, 6]	0
Bottleneck-290	[-1, 1024, 6, 6]	0
Conv2d-291	[-1, 256, 6, 6]	262,144
BatchNorm2d-292	[-1, 256, 6, 6]	512
ReLU-293	[-1, 256, 6, 6]	0
Conv2d-294	[-1, 256, 6, 6]	589,824

BatchNorm2d-295	[-1, 256, 6, 6]	512
ReLU-296	[-1, 256, 6, 6]	0
Conv2d-297	[-1, 1024, 6, 6]	262,144
BatchNorm2d-298	[-1, 1024, 6, 6]	2,048
ReLU-299	[-1, 1024, 6, 6]	0
Bottleneck-300	[-1, 1024, 6, 6]	0
Conv2d-301	[-1, 256, 6, 6]	262,144
BatchNorm2d-302	[-1, 256, 6, 6]	512
ReLU-303	[-1, 256, 6, 6]	0
Conv2d-304	[-1, 256, 6, 6]	589,824
BatchNorm2d-305	[-1, 256, 6, 6]	512
ReLU-306	[-1, 256, 6, 6]	0
Conv2d-307	[-1, 1024, 6, 6]	262,144
BatchNorm2d-308	[-1, 1024, 6, 6]	2,048
ReLU-309	[-1, 1024, 6, 6]	0
Bottleneck-310	[-1, 1024, 6, 6]	0
Conv2d-311	[-1, 256, 6, 6]	262,144
BatchNorm2d-312	[-1, 256, 6, 6]	512
ReLU-313	[-1, 256, 6, 6]	0
Conv2d-314	[-1, 256, 6, 6]	589,824
BatchNorm2d-315	[-1, 256, 6, 6]	512
ReLU-316	[-1, 256, 6, 6]	0
Conv2d-317	[-1, 1024, 6, 6]	262,144
BatchNorm2d-318	[-1, 1024, 6, 6]	2,048
ReLU-319	[-1, 1024, 6, 6]	0
Bottleneck-320	[-1, 1024, 6, 6]	0
Conv2d-321	[-1, 256, 6, 6]	262,144
BatchNorm2d-322	[-1, 256, 6, 6]	512
ReLU-323	[-1, 256, 6, 6]	0
Conv2d-324	[-1, 256, 6, 6]	589,824
BatchNorm2d-325	[-1, 256, 6, 6]	512
ReLU-326	[-1, 256, 6, 6]	0
Conv2d-327	[-1, 1024, 6, 6]	262,144
BatchNorm2d-328	[-1, 1024, 6, 6]	2,048
ReLU-329	[-1, 1024, 6, 6]	0
Bottleneck-330	[-1, 1024, 6, 6]	0
Conv2d-331	[-1, 256, 6, 6]	262,144
BatchNorm2d-332	[-1, 256, 6, 6]	512

ReLU-333	[-1, 256, 6, 6]	0
Conv2d-334	[-1, 256, 6, 6]	589,824
BatchNorm2d-335	[-1, 256, 6, 6]	512
ReLU-336	[-1, 256, 6, 6]	0
Conv2d-337	[-1, 1024, 6, 6]	262,144
BatchNorm2d-338	[-1, 1024, 6, 6]	2,048
ReLU-339	[-1, 1024, 6, 6]	0
Bottleneck-340	[-1, 1024, 6, 6]	0
Conv2d-341	[-1, 256, 6, 6]	262,144
BatchNorm2d-342	[-1, 256, 6, 6]	512
ReLU-343	[-1, 256, 6, 6]	0
Conv2d-344	[-1, 256, 6, 6]	589,824
BatchNorm2d-345	[-1, 256, 6, 6]	512
ReLU-346	[-1, 256, 6, 6]	0
Conv2d-347	[-1, 1024, 6, 6]	262,144
BatchNorm2d-348	[-1, 1024, 6, 6]	2,048
ReLU-349	[-1, 1024, 6, 6]	0
Bottleneck-350	[-1, 1024, 6, 6]	0
Conv2d-351	[-1, 256, 6, 6]	262,144
BatchNorm2d-352	[-1, 256, 6, 6]	512
ReLU-353	[-1, 256, 6, 6]	0
Conv2d-354	[-1, 256, 6, 6]	589,824
BatchNorm2d-355	[-1, 256, 6, 6]	512
ReLU-356	[-1, 256, 6, 6]	0
Conv2d-357	[-1, 1024, 6, 6]	262,144
BatchNorm2d-358	[-1, 1024, 6, 6]	2,048
ReLU-359	[-1, 1024, 6, 6]	0
Bottleneck-360	[-1, 1024, 6, 6]	0
Conv2d-361	[-1, 256, 6, 6]	262,144
BatchNorm2d-362	[-1, 256, 6, 6]	512
ReLU-363	[-1, 256, 6, 6]	0
Conv2d-364	[-1, 256, 6, 6]	589,824
BatchNorm2d-365	[-1, 256, 6, 6]	512
ReLU-366	[-1, 256, 6, 6]	0
Conv2d-367	[-1, 1024, 6, 6]	262,144
BatchNorm2d-368	[-1, 1024, 6, 6]	2,048
ReLU-369	[-1, 1024, 6, 6]	0
Bottleneck-370	[-1, 1024, 6, 6]	0

Conv2d-371	[-1, 256, 6, 6]	262,144
BatchNorm2d-372	[-1, 256, 6, 6]	512
ReLU-373	[-1, 256, 6, 6]	0
Conv2d-374	[-1, 256, 6, 6]	589,824
BatchNorm2d-375	[-1, 256, 6, 6]	512
ReLU-376	[-1, 256, 6, 6]	0
Conv2d-377	[-1, 1024, 6, 6]	262,144
BatchNorm2d-378	[-1, 1024, 6, 6]	2,048
ReLU-379	[-1, 1024, 6, 6]	0
Bottleneck-380	[-1, 1024, 6, 6]	0
Conv2d-381	[-1, 256, 6, 6]	262,144
BatchNorm2d-382	[-1, 256, 6, 6]	512
ReLU-383	[-1, 256, 6, 6]	0
Conv2d-384	[-1, 256, 6, 6]	589,824
BatchNorm2d-385	[-1, 256, 6, 6]	512
ReLU-386	[-1, 256, 6, 6]	0
Conv2d-387	[-1, 1024, 6, 6]	262,144
BatchNorm2d-388	[-1, 1024, 6, 6]	2,048
ReLU-389	[-1, 1024, 6, 6]	0
Bottleneck-390	[-1, 1024, 6, 6]	0
Conv2d-391	[-1, 256, 6, 6]	262,144
BatchNorm2d-392	[-1, 256, 6, 6]	512
ReLU-393	[-1, 256, 6, 6]	0
Conv2d-394	[-1, 256, 6, 6]	589,824
BatchNorm2d-395	[-1, 256, 6, 6]	512
ReLU-396	[-1, 256, 6, 6]	0
Conv2d-397	[-1, 1024, 6, 6]	262,144
BatchNorm2d-398	[-1, 1024, 6, 6]	2,048
ReLU-399	[-1, 1024, 6, 6]	0
Bottleneck-400	[-1, 1024, 6, 6]	0
Conv2d-401	[-1, 256, 6, 6]	262,144
BatchNorm2d-402	[-1, 256, 6, 6]	512
ReLU-403	[-1, 256, 6, 6]	0
Conv2d-404	[-1, 256, 6, 6]	589,824
BatchNorm2d-405	[-1, 256, 6, 6]	512
ReLU-406	[-1, 256, 6, 6]	0
Conv2d-407	[-1, 1024, 6, 6]	262,144
BatchNorm2d-408	[-1, 1024, 6, 6]	2,048

ReLU-409	[-1, 1024, 6, 6]	0
Bottleneck-410	[-1, 1024, 6, 6]	0
Conv2d-411	[-1, 256, 6, 6]	262,144
BatchNorm2d-412	[-1, 256, 6, 6]	512
ReLU-413	[-1, 256, 6, 6]	0
Conv2d-414	[-1, 256, 6, 6]	589,824
BatchNorm2d-415	[-1, 256, 6, 6]	512
ReLU-416	[-1, 256, 6, 6]	0
Conv2d-417	[-1, 1024, 6, 6]	262,144
BatchNorm2d-418	[-1, 1024, 6, 6]	2,048
ReLU-419	[-1, 1024, 6, 6]	0
Bottleneck-420	[-1, 1024, 6, 6]	0
Conv2d-421	[-1, 256, 6, 6]	262,144
BatchNorm2d-422	[-1, 256, 6, 6]	512
ReLU-423	[-1, 256, 6, 6]	0
Conv2d-424	[-1, 256, 6, 6]	589,824
BatchNorm2d-425	[-1, 256, 6, 6]	512
ReLU-426	[-1, 256, 6, 6]	0
Conv2d-427	[-1, 1024, 6, 6]	262,144
BatchNorm2d-428	[-1, 1024, 6, 6]	2,048
ReLU-429	[-1, 1024, 6, 6]	0
Bottleneck-430	[-1, 1024, 6, 6]	0
Conv2d-431	[-1, 256, 6, 6]	262,144
BatchNorm2d-432	[-1, 256, 6, 6]	512
ReLU-433	[-1, 256, 6, 6]	0
Conv2d-434	[-1, 256, 6, 6]	589,824
BatchNorm2d-435	[-1, 256, 6, 6]	512
ReLU-436	[-1, 256, 6, 6]	0
Conv2d-437	[-1, 1024, 6, 6]	262,144
BatchNorm2d-438	[-1, 1024, 6, 6]	2,048
ReLU-439	[-1, 1024, 6, 6]	0
Bottleneck-440	[-1, 1024, 6, 6]	0
Conv2d-441	[-1, 256, 6, 6]	262,144
BatchNorm2d-442	[-1, 256, 6, 6]	512
ReLU-443	[-1, 256, 6, 6]	0
Conv2d-444	[-1, 256, 6, 6]	589,824
BatchNorm2d-445	[-1, 256, 6, 6]	512
ReLU-446	[-1, 256, 6, 6]	0

Conv2d-447	[-1, 1024, 6, 6]	262,144
BatchNorm2d-448	[-1, 1024, 6, 6]	2,048
ReLU-449	[-1, 1024, 6, 6]	0
Bottleneck-450	[-1, 1024, 6, 6]	0
Conv2d-451	[-1, 256, 6, 6]	262,144
BatchNorm2d-452	[-1, 256, 6, 6]	512
ReLU-453	[-1, 256, 6, 6]	0
Conv2d-454	[-1, 256, 6, 6]	589,824
BatchNorm2d-455	[-1, 256, 6, 6]	512
ReLU-456	[-1, 256, 6, 6]	0
Conv2d-457	[-1, 1024, 6, 6]	262,144
BatchNorm2d-458	[-1, 1024, 6, 6]	2,048
ReLU-459	[-1, 1024, 6, 6]	0
Bottleneck-460	[-1, 1024, 6, 6]	0
Conv2d-461	[-1, 256, 6, 6]	262,144
BatchNorm2d-462	[-1, 256, 6, 6]	512
ReLU-463	[-1, 256, 6, 6]	0
Conv2d-464	[-1, 256, 6, 6]	589,824
BatchNorm2d-465	[-1, 256, 6, 6]	512
ReLU-466	[-1, 256, 6, 6]	0
Conv2d-467	[-1, 1024, 6, 6]	262,144
BatchNorm2d-468	[-1, 1024, 6, 6]	2,048
ReLU-469	[-1, 1024, 6, 6]	0
Bottleneck-470	[-1, 1024, 6, 6]	0
Conv2d-471	[-1, 256, 6, 6]	262,144
BatchNorm2d-472	[-1, 256, 6, 6]	512
ReLU-473	[-1, 256, 6, 6]	0
Conv2d-474	[-1, 256, 6, 6]	589,824
BatchNorm2d-475	[-1, 256, 6, 6]	512
ReLU-476	[-1, 256, 6, 6]	0
Conv2d-477	[-1, 1024, 6, 6]	262,144
BatchNorm2d-478	[-1, 1024, 6, 6]	2,048
ReLU-479	[-1, 1024, 6, 6]	0
Bottleneck-480	[-1, 1024, 6, 6]	0
Conv2d-481	[-1, 512, 6, 6]	524,288
BatchNorm2d-482	[-1, 512, 6, 6]	1,024
ReLU-483	[-1, 512, 6, 6]	0
Conv2d-484	[-1, 512, 3, 3]	2,359,296

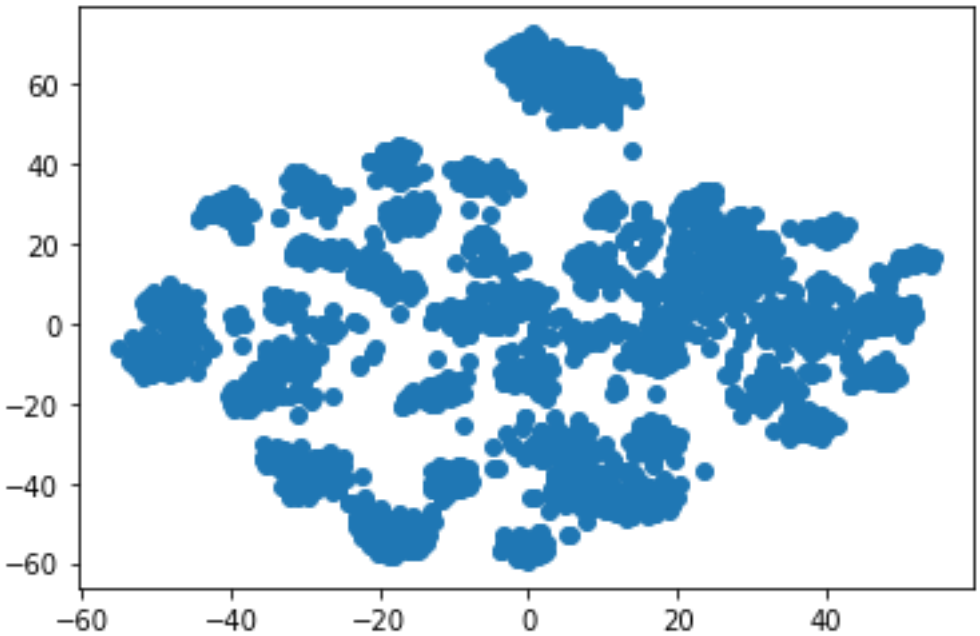
```
    BatchNorm2d-485          [-1, 512, 3, 3]         1,024
          ReLU-486          [-1, 512, 3, 3]             0
        Conv2d-487         [-1, 2048, 3, 3]      1,048,576
    BatchNorm2d-488          [-1, 2048, 3, 3]         4,096
        Conv2d-489         [-1, 2048, 3, 3]      2,097,152
    BatchNorm2d-490          [-1, 2048, 3, 3]         4,096
          ReLU-491         [-1, 2048, 3, 3]             0
     Bottleneck-492          [-1, 2048, 3, 3]             0
        Conv2d-493          [-1, 512, 3, 3]      1,048,576
    BatchNorm2d-494          [-1, 512, 3, 3]         1,024
          ReLU-495          [-1, 512, 3, 3]             0
        Conv2d-496          [-1, 512, 3, 3]      2,359,296
    BatchNorm2d-497          [-1, 512, 3, 3]         1,024
          ReLU-498          [-1, 512, 3, 3]             0
        Conv2d-499         [-1, 2048, 3, 3]      1,048,576
    BatchNorm2d-500          [-1, 2048, 3, 3]         4,096
          ReLU-501         [-1, 2048, 3, 3]             0
     Bottleneck-502          [-1, 2048, 3, 3]             0
        Conv2d-503          [-1, 512, 3, 3]      1,048,576
    BatchNorm2d-504          [-1, 512, 3, 3]         1,024
          ReLU-505          [-1, 512, 3, 3]             0
        Conv2d-506          [-1, 512, 3, 3]      2,359,296
    BatchNorm2d-507          [-1, 512, 3, 3]         1,024
          ReLU-508          [-1, 512, 3, 3]             0
        Conv2d-509         [-1, 2048, 3, 3]      1,048,576
    BatchNorm2d-510          [-1, 2048, 3, 3]         4,096
          ReLU-511         [-1, 2048, 3, 3]             0
     Bottleneck-512          [-1, 2048, 3, 3]             0
AdaptiveAvgPool2d-513          [-1, 2048, 1, 1]             0
        Linear-514                 [-1, 50]        102,450
================================================================
Total params: 58,246,258
Trainable params: 58,246,258
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.11
Forward/backward pass size (MB): 111.43
Params size (MB): 222.19
```

```
Estimated Total Size (MB): 333.72
----------------------------------------------------------------
```

2.

Accuracy on validation set: 80.56%

3.



Problem 2

1.

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1          [-1, 64, 512, 512]           1,792
              ReLU-2          [-1, 64, 512, 512]               0
            Conv2d-3          [-1, 64, 512, 512]          36,928
              ReLU-4          [-1, 64, 512, 512]               0
         MaxPool2d-5          [-1, 64, 256, 256]               0
            Conv2d-6         [-1, 128, 256, 256]          73,856
              ReLU-7         [-1, 128, 256, 256]               0
            Conv2d-8         [-1, 128, 256, 256]         147,584
```

```
            ReLU-9       [-1, 128, 256, 256]             0
       MaxPool2d-10       [-1, 128, 128, 128]             0
          Conv2d-11       [-1, 256, 128, 128]       295,168
            ReLU-12       [-1, 256, 128, 128]             0
          Conv2d-13       [-1, 256, 128, 128]       590,080
            ReLU-14       [-1, 256, 128, 128]             0
          Conv2d-15       [-1, 256, 128, 128]       590,080
            ReLU-16       [-1, 256, 128, 128]             0
       MaxPool2d-17         [-1, 256, 64, 64]             0
          Conv2d-18         [-1, 512, 64, 64]     1,180,160
            ReLU-19         [-1, 512, 64, 64]             0
          Conv2d-20         [-1, 512, 64, 64]     2,359,808
            ReLU-21         [-1, 512, 64, 64]             0
          Conv2d-22         [-1, 512, 64, 64]     2,359,808
            ReLU-23         [-1, 512, 64, 64]             0
       MaxPool2d-24         [-1, 512, 32, 32]             0
          Conv2d-25         [-1, 512, 32, 32]     2,359,808
            ReLU-26         [-1, 512, 32, 32]             0
          Conv2d-27         [-1, 512, 32, 32]     2,359,808
            ReLU-28         [-1, 512, 32, 32]             0
          Conv2d-29         [-1, 512, 32, 32]     2,359,808
            ReLU-30         [-1, 512, 32, 32]             0
       MaxPool2d-31         [-1, 512, 16, 16]             0
  ConvTranspose2d-32         [-1, 7, 512, 512]    14,680,071
================================================================
Total params: 29,394,759
Trainable params: 29,394,759
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 3.00
Forward/backward pass size (MB): 1155.00
Params size (MB): 112.13
Estimated Total Size (MB): 1270.13
----------------------------------------------------------------
```

2.

```
----------------------------------------------------------------
       Layer (type)             Output Shape        Param #
```

```
================================================================
          Conv2d-1        [-1, 64, 256, 256]          9,408
     BatchNorm2d-2        [-1, 64, 256, 256]            128
            ReLU-3        [-1, 64, 256, 256]              0
       MaxPool2d-4        [-1, 64, 128, 128]              0
          Conv2d-5        [-1, 64, 128, 128]          4,096
     BatchNorm2d-6        [-1, 64, 128, 128]            128
            ReLU-7        [-1, 64, 128, 128]              0
          Conv2d-8        [-1, 64, 128, 128]         36,864
     BatchNorm2d-9        [-1, 64, 128, 128]            128
           ReLU-10        [-1, 64, 128, 128]              0
         Conv2d-11       [-1, 256, 128, 128]         16,384
    BatchNorm2d-12       [-1, 256, 128, 128]            512
         Conv2d-13       [-1, 256, 128, 128]         16,384
    BatchNorm2d-14       [-1, 256, 128, 128]            512
           ReLU-15       [-1, 256, 128, 128]              0
     Bottleneck-16       [-1, 256, 128, 128]              0
         Conv2d-17        [-1, 64, 128, 128]         16,384
    BatchNorm2d-18        [-1, 64, 128, 128]            128
           ReLU-19        [-1, 64, 128, 128]              0
         Conv2d-20        [-1, 64, 128, 128]         36,864
    BatchNorm2d-21        [-1, 64, 128, 128]            128
           ReLU-22        [-1, 64, 128, 128]              0
         Conv2d-23       [-1, 256, 128, 128]         16,384
    BatchNorm2d-24       [-1, 256, 128, 128]            512
           ReLU-25       [-1, 256, 128, 128]              0
     Bottleneck-26       [-1, 256, 128, 128]              0
         Conv2d-27        [-1, 64, 128, 128]         16,384
    BatchNorm2d-28        [-1, 64, 128, 128]            128
           ReLU-29        [-1, 64, 128, 128]              0
         Conv2d-30        [-1, 64, 128, 128]         36,864
    BatchNorm2d-31        [-1, 64, 128, 128]            128
           ReLU-32        [-1, 64, 128, 128]              0
         Conv2d-33       [-1, 256, 128, 128]         16,384
    BatchNorm2d-34       [-1, 256, 128, 128]            512
           ReLU-35       [-1, 256, 128, 128]              0
     Bottleneck-36       [-1, 256, 128, 128]              0
         Conv2d-37       [-1, 128, 128, 128]         32,768
```

BatchNorm2d-38	[-1, 128, 128, 128]	256
ReLU-39	[-1, 128, 128, 128]	0
Conv2d-40	[-1, 128, 64, 64]	147,456
BatchNorm2d-41	[-1, 128, 64, 64]	256
ReLU-42	[-1, 128, 64, 64]	0
Conv2d-43	[-1, 512, 64, 64]	65,536
BatchNorm2d-44	[-1, 512, 64, 64]	1,024
Conv2d-45	[-1, 512, 64, 64]	131,072
BatchNorm2d-46	[-1, 512, 64, 64]	1,024
ReLU-47	[-1, 512, 64, 64]	0
Bottleneck-48	[-1, 512, 64, 64]	0
Conv2d-49	[-1, 128, 64, 64]	65,536
BatchNorm2d-50	[-1, 128, 64, 64]	256
ReLU-51	[-1, 128, 64, 64]	0
Conv2d-52	[-1, 128, 64, 64]	147,456
BatchNorm2d-53	[-1, 128, 64, 64]	256
ReLU-54	[-1, 128, 64, 64]	0
Conv2d-55	[-1, 512, 64, 64]	65,536
BatchNorm2d-56	[-1, 512, 64, 64]	1,024
ReLU-57	[-1, 512, 64, 64]	0
Bottleneck-58	[-1, 512, 64, 64]	0
Conv2d-59	[-1, 128, 64, 64]	65,536
BatchNorm2d-60	[-1, 128, 64, 64]	256
ReLU-61	[-1, 128, 64, 64]	0
Conv2d-62	[-1, 128, 64, 64]	147,456
BatchNorm2d-63	[-1, 128, 64, 64]	256
ReLU-64	[-1, 128, 64, 64]	0
Conv2d-65	[-1, 512, 64, 64]	65,536
BatchNorm2d-66	[-1, 512, 64, 64]	1,024
ReLU-67	[-1, 512, 64, 64]	0
Bottleneck-68	[-1, 512, 64, 64]	0
Conv2d-69	[-1, 128, 64, 64]	65,536
BatchNorm2d-70	[-1, 128, 64, 64]	256
ReLU-71	[-1, 128, 64, 64]	0
Conv2d-72	[-1, 128, 64, 64]	147,456
BatchNorm2d-73	[-1, 128, 64, 64]	256
ReLU-74	[-1, 128, 64, 64]	0
Conv2d-75	[-1, 512, 64, 64]	65,536

BatchNorm2d-76	[-1, 512, 64, 64]	1,024
ReLU-77	[-1, 512, 64, 64]	0
Bottleneck-78	[-1, 512, 64, 64]	0
Conv2d-79	[-1, 256, 64, 64]	131,072
BatchNorm2d-80	[-1, 256, 64, 64]	512
ReLU-81	[-1, 256, 64, 64]	0
Conv2d-82	[-1, 256, 32, 32]	589,824
BatchNorm2d-83	[-1, 256, 32, 32]	512
ReLU-84	[-1, 256, 32, 32]	0
Conv2d-85	[-1, 1024, 32, 32]	262,144
BatchNorm2d-86	[-1, 1024, 32, 32]	2,048
Conv2d-87	[-1, 1024, 32, 32]	524,288
BatchNorm2d-88	[-1, 1024, 32, 32]	2,048
ReLU-89	[-1, 1024, 32, 32]	0
Bottleneck-90	[-1, 1024, 32, 32]	0
Conv2d-91	[-1, 256, 32, 32]	262,144
BatchNorm2d-92	[-1, 256, 32, 32]	512
ReLU-93	[-1, 256, 32, 32]	0
Conv2d-94	[-1, 256, 32, 32]	589,824
BatchNorm2d-95	[-1, 256, 32, 32]	512
ReLU-96	[-1, 256, 32, 32]	0
Conv2d-97	[-1, 1024, 32, 32]	262,144
BatchNorm2d-98	[-1, 1024, 32, 32]	2,048
ReLU-99	[-1, 1024, 32, 32]	0
Bottleneck-100	[-1, 1024, 32, 32]	0
Conv2d-101	[-1, 256, 32, 32]	262,144
BatchNorm2d-102	[-1, 256, 32, 32]	512
ReLU-103	[-1, 256, 32, 32]	0
Conv2d-104	[-1, 256, 32, 32]	589,824
BatchNorm2d-105	[-1, 256, 32, 32]	512
ReLU-106	[-1, 256, 32, 32]	0
Conv2d-107	[-1, 1024, 32, 32]	262,144
BatchNorm2d-108	[-1, 1024, 32, 32]	2,048
ReLU-109	[-1, 1024, 32, 32]	0
Bottleneck-110	[-1, 1024, 32, 32]	0
Conv2d-111	[-1, 256, 32, 32]	262,144
BatchNorm2d-112	[-1, 256, 32, 32]	512
ReLU-113	[-1, 256, 32, 32]	0

Conv2d-114	[-1, 256, 32, 32]	589,824
BatchNorm2d-115	[-1, 256, 32, 32]	512
ReLU-116	[-1, 256, 32, 32]	0
Conv2d-117	[-1, 1024, 32, 32]	262,144
BatchNorm2d-118	[-1, 1024, 32, 32]	2,048
ReLU-119	[-1, 1024, 32, 32]	0
Bottleneck-120	[-1, 1024, 32, 32]	0
Conv2d-121	[-1, 256, 32, 32]	262,144
BatchNorm2d-122	[-1, 256, 32, 32]	512
ReLU-123	[-1, 256, 32, 32]	0
Conv2d-124	[-1, 256, 32, 32]	589,824
BatchNorm2d-125	[-1, 256, 32, 32]	512
ReLU-126	[-1, 256, 32, 32]	0
Conv2d-127	[-1, 1024, 32, 32]	262,144
BatchNorm2d-128	[-1, 1024, 32, 32]	2,048
ReLU-129	[-1, 1024, 32, 32]	0
Bottleneck-130	[-1, 1024, 32, 32]	0
Conv2d-131	[-1, 256, 32, 32]	262,144
BatchNorm2d-132	[-1, 256, 32, 32]	512
ReLU-133	[-1, 256, 32, 32]	0
Conv2d-134	[-1, 256, 32, 32]	589,824
BatchNorm2d-135	[-1, 256, 32, 32]	512
ReLU-136	[-1, 256, 32, 32]	0
Conv2d-137	[-1, 1024, 32, 32]	262,144
BatchNorm2d-138	[-1, 1024, 32, 32]	2,048
ReLU-139	[-1, 1024, 32, 32]	0
Bottleneck-140	[-1, 1024, 32, 32]	0
Conv2d-141	[-1, 512, 32, 32]	524,288
BatchNorm2d-142	[-1, 512, 32, 32]	1,024
ReLU-143	[-1, 512, 32, 32]	0
Conv2d-144	[-1, 512, 16, 16]	2,359,296
BatchNorm2d-145	[-1, 512, 16, 16]	1,024
ReLU-146	[-1, 512, 16, 16]	0
Conv2d-147	[-1, 2048, 16, 16]	1,048,576
BatchNorm2d-148	[-1, 2048, 16, 16]	4,096
Conv2d-149	[-1, 2048, 16, 16]	2,097,152
BatchNorm2d-150	[-1, 2048, 16, 16]	4,096
ReLU-151	[-1, 2048, 16, 16]	0

Bottleneck-152	[-1, 2048, 16, 16]	0
Conv2d-153	[-1, 512, 16, 16]	1,048,576
BatchNorm2d-154	[-1, 512, 16, 16]	1,024
ReLU-155	[-1, 512, 16, 16]	0
Conv2d-156	[-1, 512, 16, 16]	2,359,296
BatchNorm2d-157	[-1, 512, 16, 16]	1,024
ReLU-158	[-1, 512, 16, 16]	0
Conv2d-159	[-1, 2048, 16, 16]	1,048,576
BatchNorm2d-160	[-1, 2048, 16, 16]	4,096
ReLU-161	[-1, 2048, 16, 16]	0
Bottleneck-162	[-1, 2048, 16, 16]	0
Conv2d-163	[-1, 512, 16, 16]	1,048,576
BatchNorm2d-164	[-1, 512, 16, 16]	1,024
ReLU-165	[-1, 512, 16, 16]	0
Conv2d-166	[-1, 512, 16, 16]	2,359,296
BatchNorm2d-167	[-1, 512, 16, 16]	1,024
ReLU-168	[-1, 512, 16, 16]	0
Conv2d-169	[-1, 2048, 16, 16]	1,048,576
BatchNorm2d-170	[-1, 2048, 16, 16]	4,096
ReLU-171	[-1, 2048, 16, 16]	0
Bottleneck-172	[-1, 2048, 16, 16]	0
ConvTranspose2d-173	[-1, 512, 32, 32]	16,777,728
BatchNorm2d-174	[-1, 512, 32, 32]	1,024
ReLU-175	[-1, 512, 32, 32]	0
ConvTranspose2d-176	[-1, 512, 32, 32]	2,359,808
BatchNorm2d-177	[-1, 512, 32, 32]	1,024
ReLU-178	[-1, 512, 32, 32]	0
ConvTranspose2d-179	[-1, 512, 32, 32]	2,359,808
BatchNorm2d-180	[-1, 512, 32, 32]	1,024
ReLU-181	[-1, 512, 32, 32]	0
ConvTranspose2d-182	[-1, 128, 64, 64]	1,048,704
BatchNorm2d-183	[-1, 128, 64, 64]	256
ReLU-184	[-1, 128, 64, 64]	0
ConvTranspose2d-185	[-1, 128, 64, 64]	147,584
BatchNorm2d-186	[-1, 128, 64, 64]	256
ReLU-187	[-1, 128, 64, 64]	0
ConvTranspose2d-188	[-1, 128, 64, 64]	147,584
BatchNorm2d-189	[-1, 128, 64, 64]	256

```
          ReLU-190         [-1, 128, 64, 64]             0
 ConvTranspose2d-191        [-1, 32, 128, 128]         65,568
    BatchNorm2d-192        [-1, 32, 128, 128]            64
          ReLU-193        [-1, 32, 128, 128]             0
 ConvTranspose2d-194        [-1, 32, 128, 128]          9,248
    BatchNorm2d-195        [-1, 32, 128, 128]            64
          ReLU-196        [-1, 32, 128, 128]             0
 ConvTranspose2d-197        [-1, 16, 256, 256]          8,208
    BatchNorm2d-198        [-1, 16, 256, 256]            32
          ReLU-199        [-1, 16, 256, 256]             0
 ConvTranspose2d-200        [-1, 16, 256, 256]          2,320
    BatchNorm2d-201        [-1, 16, 256, 256]            32
          ReLU-202        [-1, 16, 256, 256]             0
 ConvTranspose2d-203         [-1, 8, 512, 512]          2,056
    BatchNorm2d-204         [-1, 8, 512, 512]            16
          ReLU-205         [-1, 8, 512, 512]             0
 ConvTranspose2d-206         [-1, 7, 512, 512]           511
    BatchNorm2d-207         [-1, 7, 512, 512]            14
          ReLU-208         [-1, 7, 512, 512]             0
================================================================
Total params: 46,441,221
Trainable params: 46,441,221
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 3.00
Forward/backward pass size (MB): 1731.00
Params size (MB): 177.16
Estimated Total Size (MB): 1911.16
----------------------------------------------------------------
```

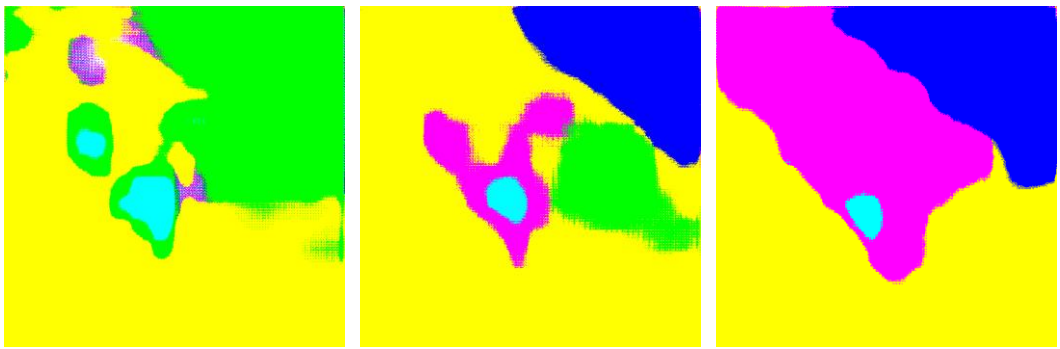
3.

mIOU on the improved model: 62.053%

4.

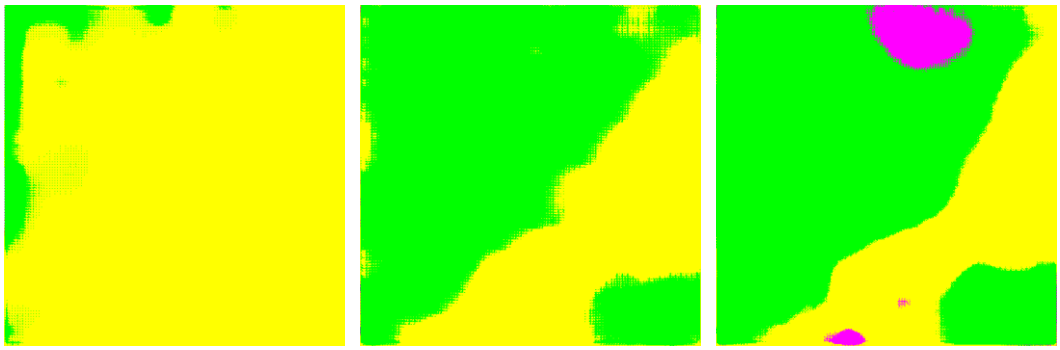
0010:

early stage -> middle stage -> final stage



0097:

early -> middle -> final



0107:

early -> middle -> final